

BENCHMARK: INACCURATE

Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2460856.963 +/- 0.002 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.172 +/- 0.029               |
| Transit depth (Rp/Rs)^2                           | 2.96 +/- 1.01 [%]             |
| Orbital Inclination (inc)                         | 89.18 +/- 0.84                |
| Airmass coefficient 1 (a1)                        | 0.869 +/- 0.011               |
| Airmass coefficient 2 (a2)                        | 0.09 +/- 0.11                 |
| Scatter in the residuals of the lightcurve fit is | 1.32 %                        |
| Best Comparison Star                              | #3 - [276, 378]               |
| Optimal Method                                    | PSF photometry                |
| Transit Duration (day)                            | 0.0598 +/- 0.0028             |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R* fit vs published    | 0.172 ± 0.029 vs 0.14762 (deviation 0.8σ) |
| Duration fit vs published | 0.0598 d vs 0.1033 d (deviation 14.79σ)   |
| Mid-time O–C              | 5.1 min                                   |
| Depth SNR                 | 5.6                                       |
| Residual scatter          | 1.32 %                                    |

Light curve

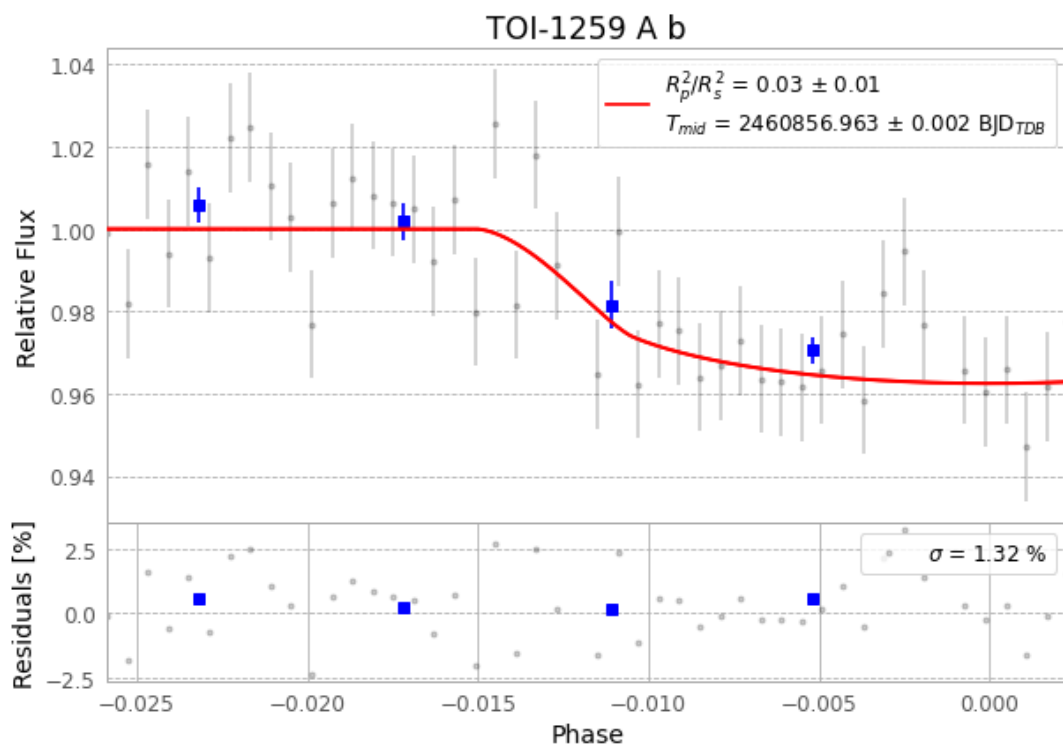


Figure 1 — FinalLightCurve\_TOI-1259 A b\_2025-06-30.png (SHA-256 0d4d4e24fd94db4c...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [389, 277], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 0df946fc3da4d599...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

48 FITS frames (32 MB), TOI1259250630085715.FITS → TOI1259250630111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1259-250630.log (bd1e4ddd17b0...), FinalLightCurve\_TOI-1259 A b\_2025-06-30.png (0d4d4e24fd94...), FinalLightCurve\_TOI-1259 A b\_2025-06-30.csv (175dc3afabcf...)

## Compute resources

Wall time 147.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:25Z.